SECTION-D

Note: Long answer type questions. Attempt any one question out of two questions. (1x10=10)

Q.19 Explain cooling curve of pure metals with diagram.

Q.20 Explain solid solubility phase with a neat diagram.

Roll No. ....

2nd Year / Advance Diploma in Tool and Die Making
Subject:- Heat Treatment

Time : 2Hrs. M.M. : 50

SECTION-A

Note: Multiple choice questions. All questions are compulsory (5x1=5)

Q.1 Which process of heat treatment used for surface hardening of steels?

a) to increase its electrical conductivity

b) to decrease its melting temperature

c) to protect from corrosion

d) to enhance its magnetic properties

Q.2 In nitriding process of heat treatment steel components are heated in presence of

a) carbon dioxide b) nitrogen

c) ammonia d) cyanide

Q.3 Hot working process:

a) increases toughness and ductility

b) increases surface finish

c) increases hardness and strength

d) none

Q.4 Annealing of metals:

a) removes internal stress

b) increases size of grains

c) decreases conductivity

d) none

Q.5 Melting point of Aluminium is

a) 660°C b) 670°C

c) 680°C d) 720°C

SECTION-B

Note: Objective type questions. All questions are compulsory. (5x1=5)

Q.6 Heat treatment process used to increase toughness and to reduce brittleness is

a) annealing b) case hardening

c) normalizing d) tempering

Q.7 Tempering is used to

a) decrease toughness b) decrease brittleness

c) decrease stiffness d) decrease resistance

Q.8 Which is solid carburizing material?

a) Charcoal b) Petrol

c) Ammonia d) Kerosene

Q.9 Cyaniding and Nitriding are two methods of

a) hardening b) case hardening

c) tempering d) normalizing

Q.10 The carbon percentage of steel does not exceed

a) 1.7% b) 0.8%

c) 1.5% d) 4.2%

SECTION-C

Note: Very Short answer type questions. Attempt any six questions out of eight questions. (6x5=30)

Q.11 Explain method of liquid carburizing with a neat sketch.

Q.12 Explain classification of steels.

Q.13 Differentiate between tempering and quenching.

Q.14 Explain any two advantages and disadvantages of tempering.

Q.15 Explain any two advantages and disadvantages of annealing.

Q.16 Explain all mechanical properties of metal.

Q.17 Explain TTT Curves.

Q.18 Explain heat treatment and its various processes.